NAÏVE BAYES CLASSIFIER

Outline

- Background
- Probability Basics
- Bayes' Theorem
- Naïve Bayes
  - Principle and Algorithm
  - Example: Play Tennis
- Relevant Issues

There are three kinds of probabiity:

1. Prior probability
2. Conditional probabilitty
3. Joint Probability

Example:
P(male), P(female) are prior or absolute probabilty

P(male|height>5.5 ft), p(female|can play football) are conditional probability

P(Weather, PlayTennis) represent various combinations, i.e., P(sunny, yes), P(sunny,no), P(overcast, yes), P(overcast, no), P(rain, yes), P(rain, no). P (Weather, PlayTennis) is a joint probability)

Probability Basics

- Prior, conditional and joint probability for random variables
  - Prior probability: $P(X)$
  - Conditional probability: $P(X_1 \mid X_2), P(X_2 \mid X_1)$
  - Joint probability: $\mathbf{X} = (X_1, X_2), P(\mathbf{X}) = P(X_1, X_2)$
  - Relationship: $P(X_1, X_2) = P(X_2 \mid X_1)P(X_1) = P(X_1 \mid X_2)P(X_2)$
  - Independence: $P(X_2 \mid X_1) = P(X_2), P(X_1 \mid X_2) = P(X_1), P(X_1, X_2) = P(X_1)P(X_2)$
- Bayes' Theorem

$$P(C \mid \mathbf{X}) = \frac{P(\mathbf{X} \mid C)P(C)}{P(\mathbf{X})}$$

$$Posterior\ \Pr obability = \frac{Likelihood \times Class\ \Pr ior\ \Pr obability}{\Pr edictior\ \Pr ior\ \Pr obability}$$

Bayes' Theorem

HIV global prior probability = 0.008

P(T|HIV) = 95%

P(~T|~HIV) = 95%

Perform a test and the result is positive
What is the probability of having HIV?

$P(HIV|T) \approx P(T|HIV) \times P(HIV) = 0.95 \times 0.008 = 0.0076$

$P(\sim HIV|T) \approx P(T|\sim HIV) \times P(\sim HIV) = 0.05 \times 0.992 = 0.0496$ (6.5x)

The sum is not 1. Why? The denominator P(T) of Bayes' theorem has been ignored.

$P(T) = P(T|HIV) \times P(HIV) + P(T|\sim HIV) \times P(\sim HIV) = P(T,HIV)+P(T,\sim HIV)$

$= P(T)$

$P(HIV|T) = 0.0076/(0.0076+0.0496) = 0.13$

$P(\sim HIV|T) = 0.0496/(0.0076+0.0496) = 0.87$

The Naïve Bayes Classifier

- The Naïve Bayes classifier is based on the Bayes' theorem with independence assumptions between predictors

- A Naïve Bayes model is easy to build, with no complicated iterative parameter estimation which makes it particularly useful to deal with very large datasets

- The Naïve Bayes classifier often does surprisingly well, outperforming more sophisticated classification methods

The Naïve Bayes Classifier

- The Naïve Bayes method assumes that the probability $P(B \wedge C \wedge D|A)$, which is difficult to compute, can instead be substituted by a "naïve" approximation that assumes, for a given class, the values of the attributes to be independent.

- This means that $P(B \wedge C \wedge D|A)$ is replaced by

$$P(B|A) \times P(C|A) \times P(D|A)$$

- Which is easy to compute, since each of these factors can be easily estimated from the table of instances.

Example

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Classify Outlook = Sunny,Temperature = High, Humidity = High and Wind = Strong

P(PlayTennis = Yes | O = S, T = H, Hm = H, W =S)
= P(O = S, T = H, Hm = H, W = S | PlayTennis = Yes) X P(Yes)
/P(O = S, T = H, Hm = H, W = S)

P(PlayTennis = No | O = S, T = H, Hm = H, W =S)
= P(O = S, T = H, Hm = H, W = S | PlayTennis = No) X P(No)
/P(O = S, T = H, Hm = H, W = S)

Denominator can be ignored, because it has equal impacts in the above two expressions.

Classify Outlook = Sunny,Temperature = High, Humidity = High and Wind = Strong

$P(\text{PlayTennis} = \text{Yes} \mid O = S, T = H, Hm = H, W = S)$
$= P(O = S, T = H, Hm = H, W = S \mid \text{PlayTennis} = \text{Yes}) \times P(\text{Yes})$

$P(\text{PlayTennis} = \text{No} \mid O = S, T = H, Hm = H, W = S)$
$= P(O = S, T = H, Hm = H, W = S \mid \text{PlayTennis} = \text{No}) \times P(\text{No})$

$P(\text{Yes} \mid O = S, T = H, Hm = H, W = S) = P(O = S \mid \text{Yes})\, P(T = h \mid \text{yes})\, P(Hm = h \mid \text{Yes})\, P(W = S \mid \text{Yes})\, P(\text{Yrs})$
$= 2/9 * 2/9*3/9*2/9 * 9/14 = 0.033$

$P(\text{No} \mid O = S, T = H, Hm = H, W = S) = P(O = S \mid \text{No})\, P(T = h \mid \text{No})\, P(Hm = h \mid \text{No})\, P(W = S \mid \text{No})\, P(\text{No})$
$= 3/5 * 2/5*4/5*2/5*5/14 = 0.027$

The prediction is Yes

Naïve Bayes

- Example: Continuous-valued Features
  - Temperature is naturally of continuous value.

    Yes: 25.2, 19.3, 18.5, 21.7, 20.1, 24.3, 22.8, 23.1, 19.8

    No: 27.3, 30.1, 17.4, 29.5, 15.1
  - Estimate mean and variance for each class

$$\mu = \frac{1}{N}\sum_{n=1}^{N} x_n, \quad \sigma^2 = \frac{1}{N}\sum_{n=1}^{N}(x_n - \mu)^2$$

$$\mu_{Yes} = 21.64, \ \sigma_{Yes} = 2.35$$

$$\mu_{No} = 23.88, \ \sigma_{No} = 7.09$$

  - **Learning Phase**: output two Gaussian models for P(temp|C)

$$\hat{P}(x \mid Yes) = \frac{1}{2.35\sqrt{2\pi}} \exp\left(-\frac{(x-21.64)^2}{2\times 2.35^2}\right) = \frac{1}{2.35\sqrt{2\pi}} \exp\left(-\frac{(x-21.64)^2}{11.09}\right)$$

$$\hat{P}(x \mid No) = \frac{1}{7.09\sqrt{2\pi}} \exp\left(-\frac{(x-23.88)^2}{2\times 7.09^2}\right) = \frac{1}{7.09\sqrt{2\pi}} \exp\left(-\frac{(x-23.88)^2}{50.25}\right)$$

$$\hat{P}(x \mid Yes) = \frac{1}{2.35\sqrt{2\pi}} \exp\left(-\frac{(x-21.64)^2}{2\times 2.35^2} \right) = \frac{1}{2.35\sqrt{2\pi}} \exp\left(-\frac{(x-21.64)^2}{11.09} \right)$$

$$\hat{P}(x \mid No) = \frac{1}{7.09\sqrt{2\pi}} \exp\left(-\frac{(x-23.88)^2}{2\times 7.09^2} \right) = \frac{1}{7.09\sqrt{2\pi}} \exp\left(-\frac{(x-23.88)^2}{50.25} \right)$$

Classsify Outlook = Sunny, Temp = 25, Humidity = High and Wind = Strong

P(Temp = 25 | yes) = 0.061

P(Temp = 25 } No) = 0.055

P(Yes |O = S, T = 25, Hm = H, W = S) = P (O = S | Yes) P(T = 25| yes) P(Hm = h | Yes) P(W = S | Yes) P(Yes)

= 2/9 * 0.061*3/9*2/9 * 9/14 = 0.0006

P(No| O = S, T = 25, Hm = H, W = S) = P (O = S | No) P(T = 25| No) P(Hm = h | No) P(W = S|No) P(No)

= 3/5 * 0.055*4/5*2/5*5/14 = 0.004

Prediction is No

Relevant Issues

- Violation of Independence Assumption
  - For many real world tasks, $P(X_1,\cdots,X_n \mid C) \neq P(X_1 \mid C)\cdots P(X_n \mid C)$
  - Nevertheless, naïve Bayes works surprisingly well anyway!
- Zero conditional probability Problem
  - If no example contains the feature value $X_j = a_{jk}$, $\hat{P}(X_j = a_{jk} \mid C = c_i) = 0$
  - In this circumstance, $\hat{P}(x_1 \mid c_i)\cdots\hat{P}(a_{jk} \mid c_i)\cdots\hat{P}(x_n \mid c_i) = 0$ during test
  - For a remedy, conditional probabilities re-estimated with

$$\hat{P}(X_j = a_{jk} \mid C = c_i) = \frac{n_c + mp}{n + m}$$

n_c : number of training examples for which $X_j = a_{jk}$ and $C = c_i$

n : number of training examples for which $C = c_i$

p : prior estimate (usually, $p = 1/t$ for t possible values of X_j)

m : weight to prior (number of "virtual" examples, $m \geq 1$)

Outlook = Sunny in 0 +ve examples P(Outlook = Sunny | Class = yes) = 0/9
Outlook = Overcast in 4 +ve examples P(Outlook = Overcast |Class = Yes) = 4/9
Outlook = Rainy in 5 +ve examples P(Outlook = Rainy | Class = Yes) = 5/9

P(Yes |O = S, T = H, Hm = H, W = S) = P (O = S | Yes) P(T = h| yes) P(Hm = h | Yes) P(W = S|Yes) P(Yes) = 0/9 * 2/9*3/9*2/9 * 9/14 = 0

P(No| O = S, T = H, Hm = H, W = S) = P (O = S | No) P(T = h| No) P(Hm = h | No) P(W = S|No) P(No) = 3/5 * 2/5*4/5*2/5*5/14 = 0.027

New estimate: Add one example for Outlook = Sunny, Outlook = Overcast and Outlook = Rainy

P(Outlook = Sunny | Class = yes) = 1/12
P(Outlook = Overcast |Class = Yes) = 5/12
P(Outlook = Rainy | Class = Yes) = 6/12
Now follow the previous procedure